LEARNING OUTCOMES

At the end of this chapter the students came to know about the following:

1. **Knowledge about basic electronic principles**
2. **Atomic structure of elements.**
3. **Classification of Elements**
4. **Detailed knowledge of Semiconductors and its working**
5. **Working of PN-junction**

GLOSSARY

S. No	Terms	Explanation
1	Acceptor atoms	Trivalent atoms that accept free electrons from pentavalent atoms
2	Atomic number	The number of positive charges or protons in the nucleus of an atom
3	Covalent bond	The way some electrons complete their valence shells by sharing valence electrons with neighbouring atoms
4	Electron	Smallest sub atomic particle of negative charge that orbits the nucleus of an atom
5	Free electron	Electrons that are not in any orbit around the nucleus
6	Hole	A gap left in the covalent band when a valence electron gains sufficient energy to jump to the conduction band
7	Junction	The area where the P- type material and N-type material meet in the semiconductor
8	Semiconductor	An element which is neither a good conductor or a good insulator, but rather lies somewhere between the two
9	Proton	Sub atomic particle within the nucleus of an atom. Has a positive charge
10	Valence shell	The outermost electron shell for a given atom. The number of electrons in this shell determines the conductivity of the atom

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